

A causal role of the NMDA receptor in recurrent processing during perceptual inference: NMDA receptor antagonist memantine selectively improves EEG decoding of the Kanizsa illusion

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This study presents a new, **fundamental** finding to the field interested in recurrent processing and its neuromodulatory underpinnings, finding unexpectedly that memantine (blocking NMDA-receptors) enhances the decoding of features thought to rely on NMDA-receptors. This interesting, **compelling** result identifies new directions for researchers studying consciousness, sensory processing, attention, and neurotransmitters.

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Abstract

Perceptual inference requires the integration of visual features through recurrent processing, the dynamic exchange of information between higher and lower level cortical regions. While animal research has demonstrated a crucial role of NMDA receptors in recurrent processing, establishing a causal link between NMDA receptors and recurrent processing in humans has remained challenging. Here, we report two pharmacological studies with randomized, double-blind, crossover designs in which we administered the NMDA antagonist memantine, while collecting human electroencephalography (EEG). We trained and tested EEG classifiers to reflect the processing of specific stimulus features with increasing levels of complexity, namely differences in stimulus contrast, collinearity between local line elements, and illusory surfaces of a Kanizsa triangle. In two experiments involving different participants and visual tasks, we found that memantine selectively improved decoding of the Kanizsa illusion, known to depend on recurrent processing, while leaving decoding of contrast and collinearity largely unaffected. Interestingly, the results from an attentional blink

(experiment 1) and task-relevance manipulation (experiment 2) showed that memantine was only effective when the stimulus was attended and consciously accessed. These findings suggest that NMDA inhibition through memantine enhances recurrent processing, especially for attended objects, and thereby provide a crucial step toward bridging animal and human research, shedding light on the neural mechanisms underpinning perceptual inference and conscious perception.

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Introduction

Different theories of consciousness converge on a central principle: consciousness entails integration of information through recurrent neural processes that exceeds the capabilities of mere feedforward processing (Mashour et al., 2020 [↗](#); Seth & Bayne, 2022 [↗](#)). Recurrent neural mechanisms encompass feedback connections between higher- and lower-order cortical areas as well as lateral connections within sensory areas. Studies in mice and monkeys have revealed a causal role of feedback from higher cortical areas to sensory areas for tactile perception (Manita et al., 2015 [↗](#); Sachidhanandam et al., 2013 [↗](#); Takahashi et al., 2016 [↗](#)), detection of odors (Allen et al., 2017 [↗](#)), and visual stimulus attributes (Ciceri et al., 2023 [↗](#); Lamme et al., 1998 [↗](#); van Vugt et al., 2018 [↗](#); Zatzka-Haas et al., 2021 [↗](#)). In macaques, general anesthetics targeted at reducing the overall level of consciousness of the animal reduce global cortical recurrent interactions (Mashour & Hudetz, 2018 [↗](#)) and block feedback connections to V1, while sparing feedforward processing (Lamme et al., 1998 [↗](#)). Although in humans indirect evidence for a role of recurrent processing in consciousness and perceptual integration has been provided, for example by focusing on neural markers of figure-ground segregation or backward masking (Fahrenfort et al., 2007 [↗](#), 2008 [↗](#), 2012 [↗](#), 2017 [↗](#); Kok et al., 2016 [↗](#); Kok & de Lange, 2014 [↗](#); Noorman et al., 2023 [↗](#); Scholte et al., 2008 [↗](#); Wyatte et al., 2014 [↗](#)), the underlying biochemistry has remained so far elusive. In this study, we aimed to provide direct evidence for a causal role of NMDA receptors in recurrent processing (through lateral and feedback connections) and their role in perceptual inference and conscious access, factoring in potential modulating influences of attention (Flohr et al., 1998 [↗](#); Lamme, 2010 [↗](#); Sachidhanandam et al., 2013 [↗](#); Self et al., 2012 [↗](#); Thiele, 2012 [↗](#); Thiele & Bellgrove, 2018 [↗](#); Van Kerkoerle et al., 2014 [↗](#); Wang, 2008 [↗](#)).

NMDA has a slower time constant than most other receptors (decay time constant ~100 ms, voltage-gated) and is known to be densely populated in recurrent connections in the cortex (Wang, 2008 [↗](#)). In macaque monkeys, blocking NMDA receptors selectively disrupted late onset modulations of neural activity in V1 (i.e., recurrent processing) during figure-ground detection, whereas blocking AMPA receptors did not, but instead reduced the strength of the first feedforward sweep (Self et al., 2012 [↗](#); Van Kerkoerle et al., 2014 [↗](#)). This suggests that feedback connections target neuronal synapses that are rich in NMDA receptors (among other receptors), whereas feedforward connections target synapses especially rich in AMPA receptors (Thiele, 2012 [↗](#); Thiele & Bellgrove, 2018 [↗](#)). In mice, blocking NMDA receptors strongly reduced hit rates for the detection of whisker movements (Sachidhanandam et al., 2013 [↗](#)). In humans, few studies have investigated the role of NMDA receptors in recurrent processing and (conscious) perception. Blocking NMDA receptors via administration of ketamine resulted in impaired feature integration in a texture discrimination task (Meuwese et al., 2013 [↗](#)) and a distortion of the neural representations of objects in early visual cortex (van Loon et al., 2016 [↗](#)). However, as these human studies did not clearly distinguish between markers of feedforward and recurrent processing, they could not determine whether these effects were specific to recurrent processing or reflected other non-specific effects. Ketamine has rather broad adverse side effects, such as sedation and dissociation, which could potentially explain the observed behavioral deficits and neural distortions. Also, ketamine is known for its complex pharmacological dynamics, having

effects on several neurotransmitter systems, including catecholaminergic, cholinergic, nicotinic, muscarinic and GABAergic receptors (Kocsis et al., 2013 [↗](#); Mathew et al., 2012 [↗](#); Müller et al., 2011 [↗](#)).

In this study, we used a novel Kanizsa stimulus whose different visual features were designed to target distinct processing stages with increasing levels of complexity, namely local contrast differences, differences in collinearity of local line elements and the presence/absence of an illusory triangle (**Fig. 1A** [↗](#)) (see also Noorman et al., 2023 [↗](#)). The processing of collinearity of line elements predominantly relies on lateral connections (Bosking et al., 1997 [↗](#); Gilbert & Wiesel, 1979 [↗](#); Li, 1998 [↗](#); Liang et al., 2017 [↗](#); Schmidt et al., 1997 [↗](#); Stettler et al., 2002 [↗](#)), while processing of the illusory surface of the Kanizsa illusion relies on feedback connections (Halgren et al., 2003 [↗](#); Kok et al., 2016 [↗](#); Kok & de Lange, 2014 [↗](#); Lee & Nguyen, 2001 [↗](#); Pak et al., 2020 [↗](#); Wokke et al., 2013 [↗](#)). Therefore, multivariate EEG decoding of the presence/absence of the illusory Kanizsa triangle served as a marker of feedback processing (Fahrenfort et al., 2017 [↗](#); Noorman et al., 2023 [↗](#)). The ancillary features of the stimulus, the two-legged white circles, could be aligned to form a non-illusory triangle, enabling us to use collinearity decoding as a marker of lateral processing. By rotating the Kanizsa by 180 degrees, thus either having one black Pacman on top of the screen or two, we created differences in contrast between the upper and lower visual field. Because local contrast differences are processed early in the visual system, this allowed us to employ local contrast decoding as a marker of feedforward processing. In recent work we have validated that the decoding profiles of these features of different complexities at different points in time, in combination with the associated topography, can indeed serve as EEG markers of feedforward, lateral and feedback processes (Fahrenfort et al., 2017 [↗](#); Noorman et al., 2023 [↗](#)).

We tested how these neural markers were affected by memantine, a low-affinity, voltage-dependent, uncompetitive NMDA antagonist. Memantine is well tolerated and has minimal side effects compared to ketamine.

We performed two separate experiments. In experiment 1, we perturbed conscious access through masking and inattention using a modified attentional -blink (AB) paradigm (**Fig. 1B** [↗](#)). In experiment 2, we manipulated whether the illusion, collinearity, or local contrast were task-relevant or not. We hypothesized that disrupting the reentrant glutamate signal via blocking NMDA receptors by memantine would impair illusion and possibly collinearity decoding, as putative markers of feedback and lateral processing, but would spare the decoding of local contrast differences, our marker of feedforward processing. To foreshadow our results, memantine indeed specifically affected illusion decoding, but enhancing rather than impairing it. In the Discussion, we offer explanations for this surprising finding, including the effect of memantine on extrasynaptic NMDA receptors in GABAergic cells, which may have resulted in boosted visual activity.

Results

Memantine caused minimal side effects

The study had a randomized, double-blind, crossover design in which a placebo was administered during one session and memantine (20 mg) during another. Drug order was counterbalanced across the 29 healthy participants. Potential side effects of memantine on different aspects of the participant's subjective state were assessed with the visual analog scale (VAS) (Bond & Lader, 1974 [↗](#); Danion et al., 1989 [↗](#)). The only significant effect was a moderate decrease in the category “calmness” for memantine compared to placebo ($t_{28}=-23.07$, $P=0.005$, $BF_{10}=8.71$; **Fig. 1C** [↗](#), for

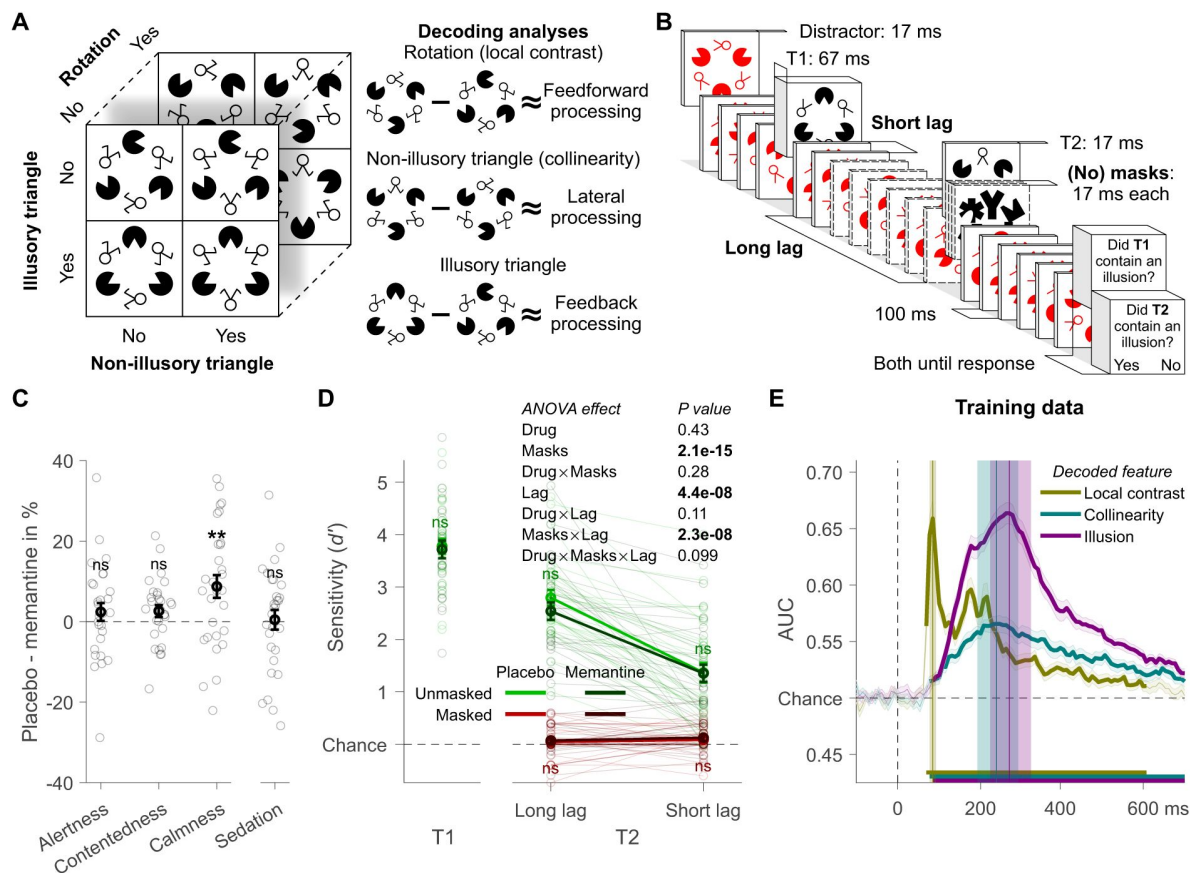


Figure 1.

Experiment 1: design, behavior, and training data.

(A) Target stimulus set and schematic of the markers for the different types of processing (showing example stimuli). (B) Trial design. (C) Effects of memantine on subjective state assessed with the visual analogue scale. The categories alertness, contentedness, and calmness are based on Bond and Lader (1974) [\[1\]](#), sedation on Danion et al. (1989) [\[2\]](#). For details see Methods. (D) Participants' ability to detect the Kanizsa illusion in the second target stimulus (T2). (E) Tenfold cross-validation decoding scheme applied to an independent training dataset. Mean decoding performance, area under the receiver operating characteristic curve (AUC), over time $\pm$ standard error of the mean (SEM) is shown. Thick lines differ from chance: $P < 0.05$, cluster-based permutation test. Local contrast, collinearity, and illusion decoding accuracy peaked at 86, 242, and 273 ms after stimulus onset and these peaks were encompassed by a 78–94, 192–292, and 223–323 ms time window, respectively. Error bars are mean $\pm$ SEM. Individual data points are plotted using low contrast. Ns is not significant ($P > 0.05$). ** $P = 0.005$.

details see Methods). Note that in experiment 2 ([Fig. 4](#)) none of these subjective measures differed between memantine and placebo, indicating that overall subjective side effects of memantine were minimal.

Memantine selectively improves decoding of the Kanizsa illusion

We recorded EEG signals from participants who detected illusory surfaces in two black targets (T1 and T2) amongst red distractors in a rapid serial visual presentation (RSVP) task ([Fig. 1B](#)). Visibility of T2 was manipulated in two ways, resulting in a 2×2 factorial task design. First, T2 was either masked or unmasked. Second, shortening the interval (lag) between the two targets from 900 to 200/300 ms induced an attentional blink (AB), which reduced T2 visibility as well. At the end of a trial, participants indicated for each target whether they had perceived an illusory surface or not. For all main analyses of behavior and EEG decoding we conducted repeated-measures ANOVAs with the factors drug (memantine vs. placebo), T1-T2 lag (short, inducing an AB, vs. long, no AB), and masks (masked vs. unmasked). The complete results of these ANOVAs are presented in the corresponding results figures. In the text, we only highlight the key results and report selectively additional statistics such as post - hoc tests.

As expected, both masking and the AB impaired participant's ability to detect the Kanizsa illusion in the second target (T2). Our masking procedure was highly effective, leading to poor T2 performance on masked trials (red lines in [Fig. 1D](#)). The AB was also effective, leading to a sharp drop in T2 performance on short lag versus long lag trials (green lines in [Fig. 1D](#)). Memantine had however no effect on perceptual sensitivity ([Fig. 1D](#)).

To establish our markers for the different neural processes of interest, the processing of different visual features (local contrast, collinearity, and the Kanizsa illusion) were decoded ([Fig. 1A](#)). For these decoding analyses, classifiers were trained using an independent training set obtained from a separate experimental task where the same stimuli were presented, but outside the context of the RSVP task (see Methods for details). No drugs were administered during this separate “localizer” task. In brief, during the localizer, participants had to match the target stimulus to different example stimuli (“templates”). For successful performance in this template-matching task, all three features of the stimulus had to be attended, and all features were thus task-relevant. We also used the independent training set from this template-matching task to determine the time windows for local contrast, collinearity, and illusion decoding. First, we decoded each visual feature from the localizer using a tenfold cross-validation scheme by training linear discriminant classifiers to categorize the feature as either present or absent based on the raw EEG activity across all electrodes. After testing the classifiers, each decoded visual feature showed a clear peak in classifier performance (area under the receiver operating characteristic curve [AUC]) ([Hand & Till, 2001](#)) over time: local contrast at 86 ms after stimulus onset, collinearity at 242 ms, and the illusion at 273 ms. In the following, we refer to “time window-based analyses” when we focus on the averages of the time windows encompassing these peaks (for the exact time windows used see [Fig. 1E](#)). The timing and mostly occipital nature (see the covariance/class separability maps of [Fig. 2A](#)) ([Haufe et al., 2014](#)) of these windows were consistent with the neural processes they are thought to reflect and in line with our previous EEG decoding studies ([Fahrenfort et al., 2017](#); [Noorman et al., 2023](#)). Because of this topology, subsequent decoding analyses were restricted to occipitoparietal electrodes. To increase power, we averaged each time window in the localizer task's raw EEG data. Classifiers were trained on these averages and tested on all time samples of each condition of the main RSVP task's data (see Methods) ([King & Dehaene, 2014](#)). Note that this approach therefore reflects “off-diagonal decoding” in the generalization across time approach. We show all “on-diagonal” analyses in the [supplementary figures S2-S4](#), which revealed very similar results.

We first examined the effect of memantine on the decoding of the different features of the first target (T1) from the RSVP task. For each time sample, we tested whether the memantine conditions differed in classifier performance from the placebo conditions. Cluster-based permutation testing

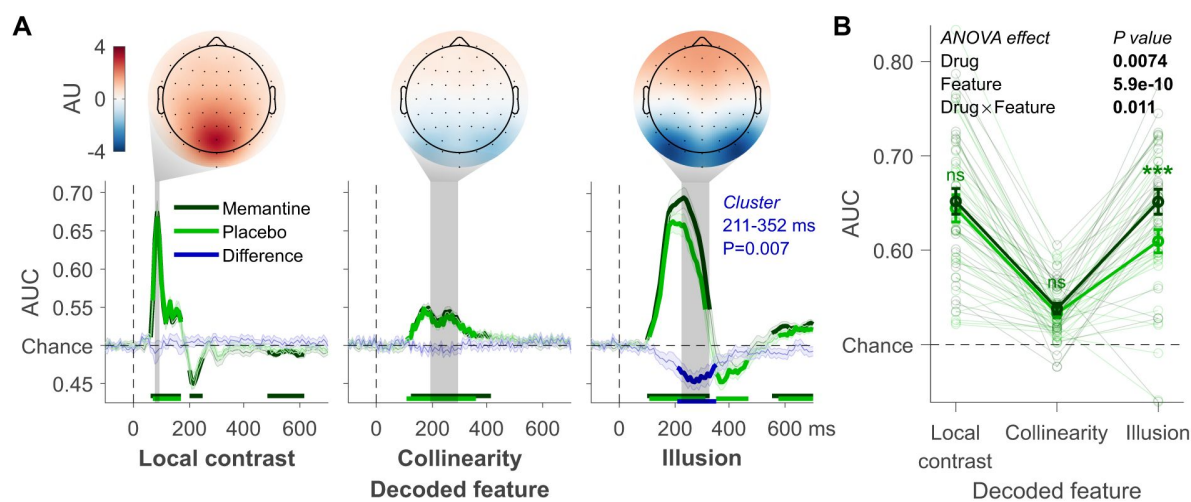


Figure 2.

Decoding applied to the T1 data.

(A) Local contrast, collinearity, and Kanizsa illusion decoding. These three visual features' time windows: 78 -94, 192-292, and 223-323 ms, respectively. Covariance/class separability maps with arbitrary units (AU) reflecting underlying neural sources are shown. Below these maps: mean decoding performance, area under the receiver operating characteristic curve (AUC), over time $\pm$ standard error of the mean (SEM). Thick lines differ from chance: $P < 0.05$, cluster-based permutation test. (B) Mean AUC for every time window. Error bars are mean $\pm$ SEM. Individual data points are plotted using low contrast. Ns is not significant ($P > 0.05$). *** $P < 0.001$.

was used to correct for multiple comparisons (Maris & Oostenveld, 2007). Crucially, only illusion decoding showed a significant cluster ranging from 211 to 352 ms after stimulus onset (Fig. 2A, blue lines). A similar result was obtained from the time window-based analysis (Fig. 2B, ANOVA results in the figure). Memantine significantly improved overall T1 decoding ($F_{1,28}=8.33$, $P=0.007$) and drug interacted with visual feature ($F_{1,28}=5.16$, $P=0.011$). Post hoc t -tests confirmed that memantine selectively affected illusion decoding accuracy ($t_{28}=-4.14$, $P<0.001$, $BF_{10}=101.92$), thus leaving local contrast ($t_{28}=-0.60$, $P=0.560$, $BF_{10}=0.23$) and collinearity decoding unaffected ($t_{28}=-1.04$, $P=0.306$, $BF_{10}=0.32$). As memantine is an NMDA blocker, we had initially hypothesized that it would impair recurrent processing, and therefore decrease illusion decoding accuracy. Surprisingly however, although memantine specifically affected illusion decoding, it *increased* instead of decreased illusion decoding. Recent studies using memantine have however mostly observed increases in task performance and/or neural markers of sensory processing (Korostenskaja et al., 2007; Light et al., 2017; Szwedlow et al., 2016, 2020), and are therefore mostly in line with our observations here. To foreshadow our further results, this effect was robust as we obtained the same *increase* in Kanizsa decoding for T2s in this study (Fig. 3), and the effect was replicated in study 2 (Fig. 5). We address this issue further in the discussion.

The role of consciousness and attention in memantine's effect on illusion decoding

To determine how memantine interacted with our manipulations of consciousness (masking) and attention (T1-T2 lag, i.e., attentional blink [AB]), we examined the decoding of the different features of the second target (T2) from the RSVP task (Fig. 3A, C, and E). First, we tested whether masking and the AB affected local contrast, collinearity and illusion decoding in the way we expected. Indeed, local contrast decoding was unaffected by the AB (Fig. 3B), and although it was impaired by masking, the drop in decoding accuracy was numerically small (0.012 AUC on average) and considerably smaller than the effect of masking on collinearity (0.026 AUC) and illusion (0.123 AUC) decoding. These results are in line with theoretical proposals and empirical findings that suggest limited effects of masking and attentional manipulations on feedforward processes (Dehaene et al., 2006; Fahrenfort et al., 2007, 2017; Lamme, 2010; Lamme et al., 2002). As would be expected for markers of lateral and feedback processing, both collinearity and illusion decoding were abolished by masking (Fig. 3C-F). The AB exclusively impaired illusion decoding, but decoding accuracy for the illusion remained well above chance during the AB (Fig. 3E-F).

Similar to T1, also for T2 decoding, memantine increased illusion decoding, but only when T2 was presented outside the AB and when it was unmasked (Fig. 3E, top left, note that masking fully abolished illusion decoding). Also our time window-based analyses showed a significant effect of memantine only when the illusion was both unmasked and presented outside the AB ($t_{28}=2.76$, $P=0.010$, $BF_{10}=4.53$; Fig. 3F). Note, however, that although these post-hoc tests of the effect of memantine on illusion decoding were significant, for our time window-based analyses we did not obtain a statistically significant interaction between the AB and memantine. While the interaction between masking and memantine only approached significance ($P=0.068$), the absence of an effect of memantine in the masked condition could reflect a floor effect, given that illusion decoding in the masked condition was not significantly better than chance. Thus, although these memantine effects were slightly less robust than for T1, probably due to reduced trial counts, these results point to (but do not conclusively demonstrate) a selective effect of memantine on illusion-related feedback processing that depends on the availability of attention. In addition to the lack of the interaction effect, another potential concern with this interpretation is that in the RSVP task, the illusion was always task-relevant and hence attended, whereas local contrast and collinearity were never task-relevant. It is therefore an open question whether local contrast and collinearity decoding were unaffected by memantine, because these visual features were unattended (not task-relevant) and/or because memantine exclusively targets feedback processes. To alleviate this

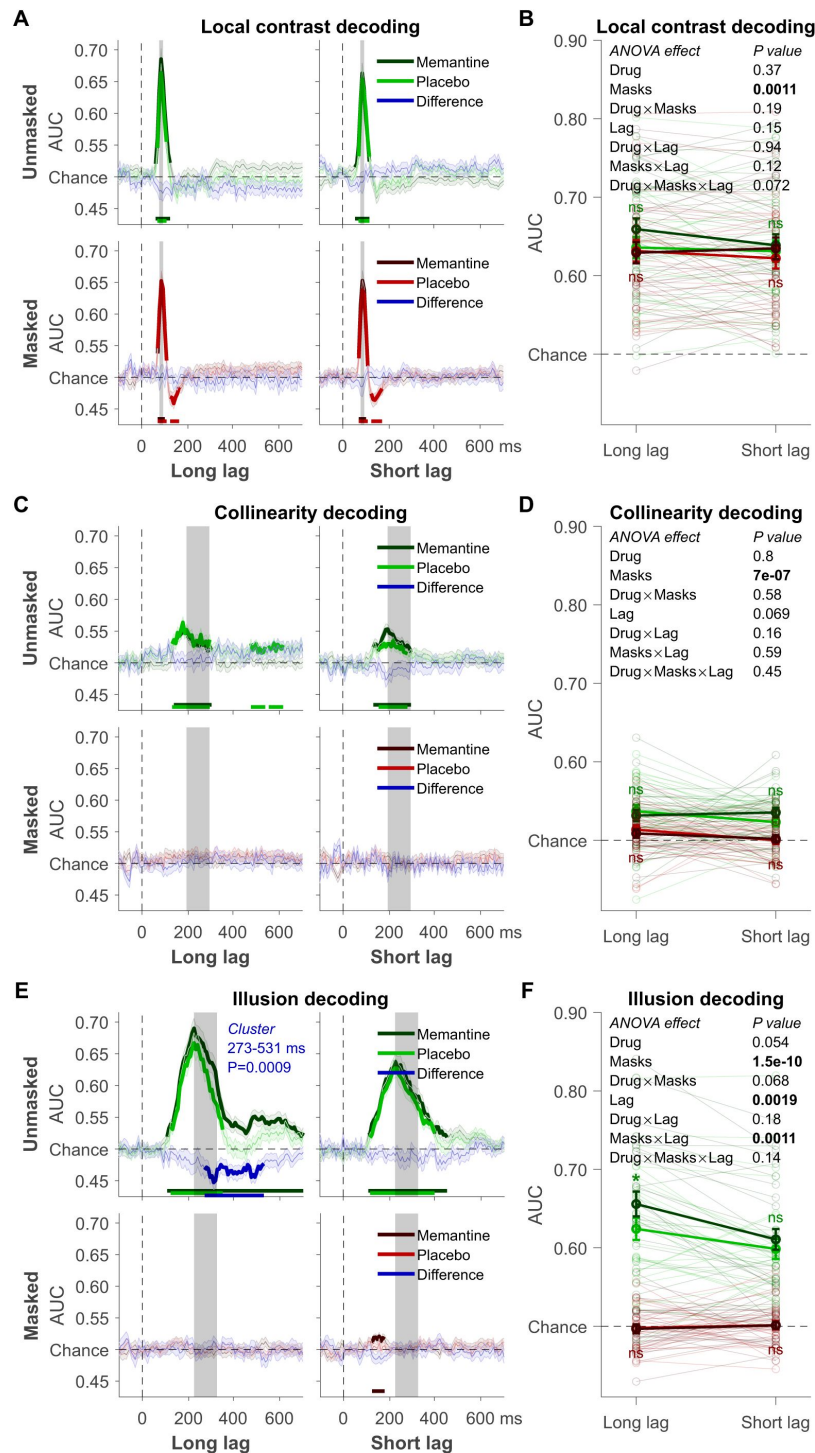


Figure 3.

Decoding of the T2 data.

(A) Local contrast decoding: mean decoding performance, area under the receiver operating characteristic curve (AUC), over time $\pm$ standard error of the mean (SEM). Thick lines differ from chance: $P < 0.05$, cluster-based permutation test. (B) Mean AUC for local contrast (rotation) decoding's time window: 78-94 ms. Error bars are mean $\pm$ SEM. Individual data points are plotted using low contrast. Ns is not significant ($P > 0.05$). (C) Collinearity decoding: mean AUC over time. (D) Mean AUC for collinearity decoding's time window: 192-292 ms. (E) Kanizsa illusion decoding: mean AUC over time. (F) Mean AUC for Kanizsa illusion's time window: 223-323 ms. * $P = 0.010$.

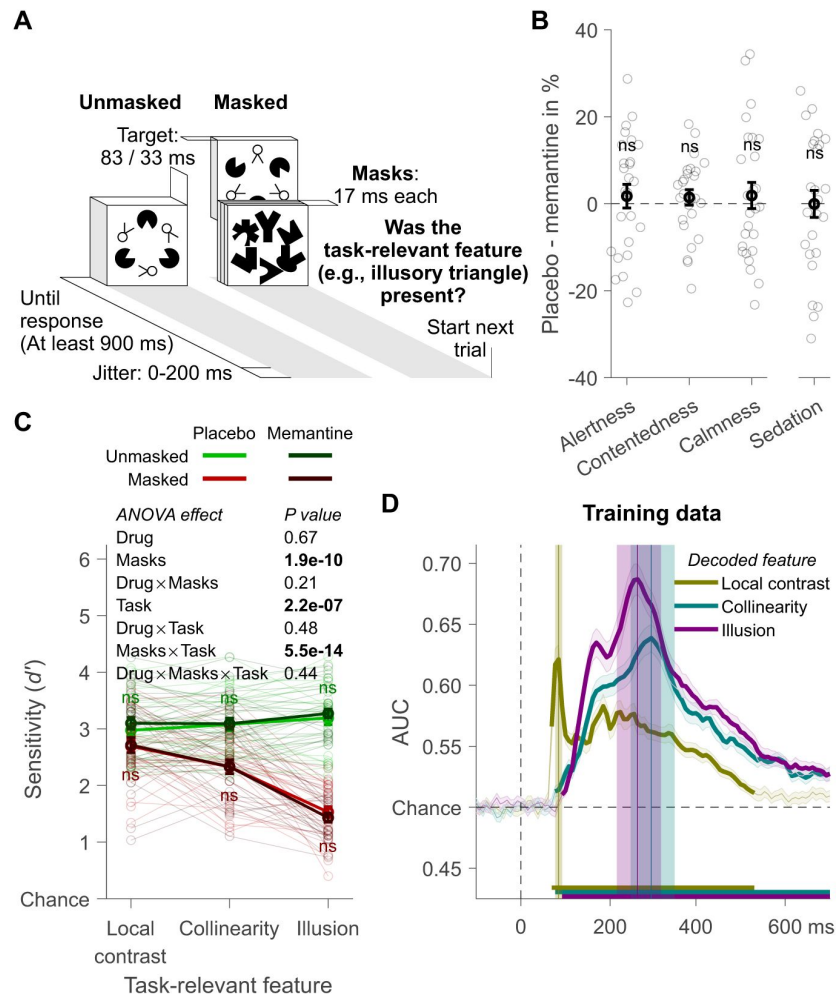


Figure 4.

Experiment 2: design, behavior, and training data.

(A) Trial design. (B) Effects of memantine on subjective state assessed with the visual analogue scale. The categories alertness, contentedness, and calmness are based on Bond and Lader (1974) [\[1\]](#), sedation on Danion et al. (1989) [\[2\]](#). For details see Methods. (C) Participants' ability to detect the task-relevant feature. (D) Tenfold cross-validation decoding scheme applied to an independent training dataset. Mean decoding performance, area under the receiver operating characteristic curve (AUC), over time $\pm$ standard error of the mean (SEM) is shown. Thick lines differ from chance: $P < 0.05$, cluster-based permutation test. Local contrast, collinearity, and illusion decoding accuracy peaked at 86, 297, and 266 ms after stimulus onset and these peaks were encompassed by a 78-94, 247-347, and 217-317 ms time window, respectively. Error bars are mean $\pm$ SEM. Individual data points are plotted using low contrast. Ns is not significant ($P > 0.05$).

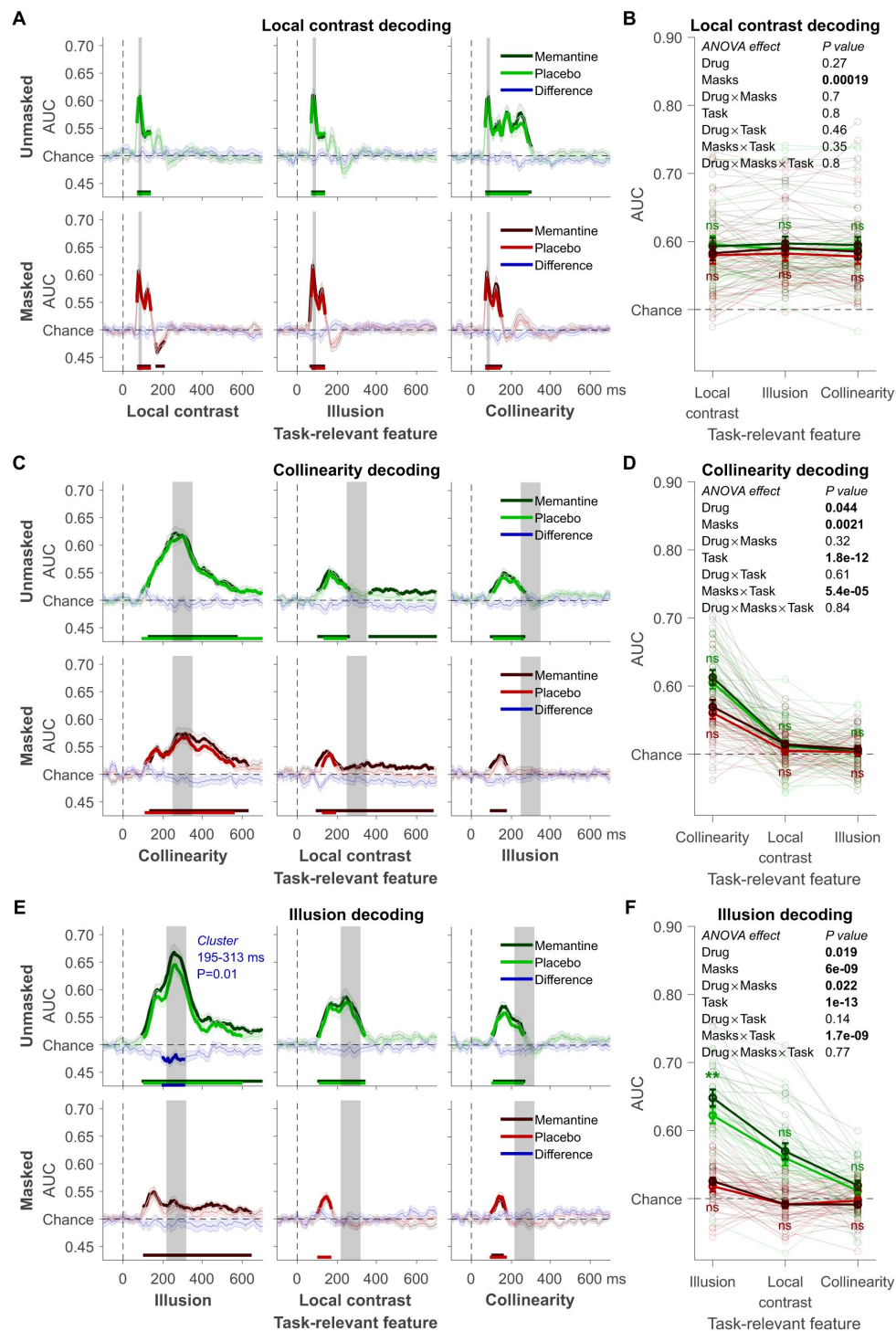


Figure 5.

Decoding results of experiment 2.

(A) Local contrast decoding: mean decoding performance, area under the receiver operating characteristic curve (AUC), over time $\pm$ standard error of the mean (SEM). Thick lines differ from chance: $P < 0.05$, cluster-based permutation test. (B) Mean AUC for local contrast decoding's time window: 78 -94 ms. Error bars are mean $\pm$ SEM. Individual data points are plotted using low contrast. Ns is not significant ($P > 0.05$). (C) Collinearity decoding: mean AUC over time. (D) Mean AUC for collinearity decoding's time window: 217-317 ms. (E) Kanizsa illusion decoding: mean AUC over time. (F) Mean AUC for Kanizsa illusion's time window: 223 -323 ms. ****** $P = 0.002$.

potential concern, and to replicate the surprising result of increased illusion decoding under memantine, we carried out a second experiment. A new group of participants was recruited and we systematically varied task-relevance of local contrast, collinearity, and the illusion in this task. This allowed us to determine whether memantine indeed specifically affected feedback processing.

Experiment 2 replicates memantine's selective effect on illusion decoding

In experiment 2, the pharmacological design was the same as before and VAS measurements now revealed no side effects from memantine (Fig. 4B). Twenty-seven new participants performed a discrimination task in which each visual feature, in separate blocks, was task-relevant (Fig. 4A). Approximately every 900-1100 ms an individual target stimulus (no RSVP) was presented for 83 ms when it was unmasked, or for 33 ms when it was masked. The timing of the masked condition was increased compared to experiment 1 to circumvent floor or ceiling effects. Performance on the Kanizsa illusion was staircased to 75% correct in a practice session by varying mask contrast, and these mask settings were used for all features (see Methods). For each target, participants indicated whether the current task-relevant feature (e.g., the Kanizsa illusion) was present or absent. In experiment 2, each visual feature was task-relevant and thus attended in some blocks, while it was task-irrelevant and thus less attended in other blocks. This allowed us to directly examine whether memantine's effects were specific to feedback processes or dependent on task-relevance of the stimulus features. For all main analyses of behavior and EEG decoding we conducted repeated-measures ANOVAs with the factors drug (memantine vs. placebo), masking (masked vs. unmasked), and task-relevant feature (local contrast, collinearity, illusion). The complete results of these ANOVAs are presented in the corresponding results figures. We found that masking impaired perceptual sensitivity, most prominently for the detection of the Kanizsa illusion, and less so for collinearity and rotation. Note that we employed less strong masking in this experiment than in experiment 1, leading to overall above chance perceptual sensitivity for the masked trials. For unmasked trials performance was at ceiling for all visual features (no differences in performance). Perceptual sensitivity was unaffected by memantine (Fig. 4C).

The EEG decoding schemes were the same as for experiment 1 and similar time windows were obtained from the independent training dataset (Fig. 4D). The localizer task was similar to the main experiment, but no masking was used. This localizer task was performed in all three sessions (intake, and two main session involving the pharmacological manipulation, 1728 trials per session). In the intake session, no drugs were administered during the localizer task. In the other two sessions, the localizer task was performed right after pill intake and this was long before memantine reached peak plasma levels (after ~4 hrs.). Overall, the timing of the decoding of the three different features in the main experiment replicated our findings from experiment 1 (Fig. 5). For local contrast, task-relevance had no effect and the significant drop in decoding accuracy caused by masking was again numerically small (0.010 AUC on average); smaller than for collinearity (0.015 AUC) and illusion decoding (0.069 AUC). Collinearity (Fig. 5C-D) and illusion (Fig. 5E-F) decoding were impaired by masking and boosted when the corresponding feature was task-relevant. Crucially, our cluster-based analyses again revealed that memantine significantly improved decoding only for the illusion and only when the illusion was unmasked and task-relevant (Fig. 5E, top left), thus replicating the results from experiment 1. Also the time window-based analysis revealed an overall main effect of drug for illusion decoding, and post-hoc test revealed that the drug effect was only significant when the illusion was unmasked and task-relevant (post hoc *t*-tests comparing memantine vs. placebo, $t_{28} = -3.36$, $P = 0.002$, $BF_{10} = 15.66$; Fig. 5F). For our time window-based analyses of illusion decoding, the specificity of the memantine effect to the unmasked condition was supported by a significant interaction between drug and masking (note, however, given overall much lower decoding accuracy in the masked condition, the lack of a memantine effect could reflect a floor effect). However, we did not obtain a statistically significant interaction between memantine and task-relevance. Thus, although the memantine

effect was significant only when the illusion was unmasked and task-relevant, just like for the effect of temporal attention in experiment 1, these results do not conclusively demonstrate a selective effect of memantine that depends attention (task-relevance). Our time window-based ANOVAs now also revealed a main effect of memantine on collinearity decoding (**Fig. 5D**), in line with the hypothesized role of NMDA receptors in recurrent (feedback and lateral) processing.

Discussion

In this study, we investigated the impact of NMDA receptor blockade on multivariate EEG decoding of different features of a visual stimulus (Kanizsa illusion, collinearity, and local contrast). We used the NMDA receptor antagonist memantine to selectively target recurrent processing while minimizing non-specific side effects, a notable improvement over previous studies blocking NMDA receptors using ketamine. Two separate experiments demonstrated selective effects of memantine on decoding accuracy of the Kanizsa illusion, our marker of feedback processing, while leaving collinearity and contrast decoding relatively unaffected, our markers for lateral and feedforward processing. Interestingly, in both experiments the effect of memantine on illusion decoding tended to be specific to attended, task-relevant stimuli presented without a backward mask. In experiment 1, the illusion was task-relevant and the effect was only found for stimuli presented unmasked and outside the attentional blink (AB), when full attention was available. In experiment 2, we obtained the effect only when the stimulus was unmasked and the illusion was task-relevant. Notably, rather than impairing illusion decoding, in both experiments memantine improved illusion decoding. These findings provide new insight into the role of NMDA-receptor mediated recurrent processing in human visual perception and will be further discussed below.

Our approach relied on using EEG decoding of different stimulus features at different points in time, together with their topography, as markers of distinct neural processes. Although such non-invasive, indirect measures of neural activity cannot provide direct evidence for feedforward vs. recurrent processes, the timing, topography, and susceptibility to masking of the decoding signatures obtained in the present study are consistent with neurophysiology (e.g., Bosking et al., 1997; Kandel et al., 2000; Lamme & Roelfsema, 2000; Lee & Nguyen, 2001; Liang et al., 2017; Pak et al., 2020), as well as with our previous work (Fahrenfort et al., 2017; Noorman et al., 2023). The specificity of the effect of memantine our EEG marker of illusion-related feedback processing is broadly consistent with previous studies indicating that ketamine, another NMDA receptor antagonist, affects perception and neural coding in visual tasks thought to rely on recurrent mechanisms (Meuwese et al., 2013; van Loon et al., 2016). The effect of ketamine in these studies likely affected intra-area lateral connections in early visual cortex (Meuwese et al., 2013) and inter-area feedback mechanisms (van Loon et al., 2016), respectively. The present effect of memantine was largely specific to illusion decoding, our marker of feedback processing, while collinearity decoding, our marker of lateral processing, was not (experiment 1) or only weakly (experiment 2) affected by memantine. We have no straightforward explanation for why NMDA receptor blockade would impact inter-areal feedback connections more strongly than intra-areal lateral connections, considering their strong functional interdependency and interaction in grouping and segmentation processes (Liang et al., 2017). One possibility is that this finding reflects properties of our EEG decoding markers for feedback vs. lateral processing: for example, decoding of the Kanizsa illusion may have been more sensitive to the relatively subtle effect of our pharmacological manipulation, either because overall decoding was better than for collinearity or because NMDA receptor dependent recurrent processes more strongly contribute to illusion decoding than to collinearity decoding.

While the effect of memantine in our study was largely specific to illusion decoding, the direction of this effect was in the opposite direction of what we had initially predicted and of the effect of ketamine in previous studies on perceptual representation: whereas ketamine impaired perceptual organization and visual cortex representations (Meuwese et al., 2013; van Loon et al.,

2016), memantine improved illusion decoding. Although both drugs are known to inhibit NMDA receptors by occupying the receptor's ion channel and are thereby blocking current flow (Glasgow et al., 2017; Molina et al., 2020), the drugs have different actions at receptors other than NMDA, with ketamine acting on dopamine D2 and serotonin 5-HT2 receptors, and memantine inhibiting several subtypes of the acetylcholine (ACh) receptor as well as serotonin 5-HT3 receptors. Memantine and ketamine are also known to target different NMDA receptor subpopulations, with their inhibitory action displaying different time courses and intensity (Glasgow et al., 2017; Johnson et al., 2015). Blockade of different NMDA receptor subpopulations can result in markedly different and even opposite results. For example, Self and colleagues (2012) found overall reduced or elevated visual activity after microinfusion of two different selective NMDA receptor antagonists (2-amino-5-phosphonovalerate and ifendipril) in macaque primary visual cortex. Although both drugs impaired the feedback-related response to figure vs. ground, similar to the effects of ketamine (Meuwese et al., 2013; van Loon et al., 2016) such opposite effects on overall activity demonstrate that the effects of NMDA antagonism strongly depend on the targeted receptor subpopulation, each with distinct functional properties.

Indeed, previous studies have also found opposite effects of memantine and ketamine. For example, while ketamine diminishes the amplitude of the so-called mismatch negativity (MMN) EEG response to (auditory) oddball stimuli (Rosburg & Kreitschmann-Andermahr, 2016), a single dose of memantine enhances MMN amplitude (Korostenskaja et al., 2007; Light et al., 2017; Swerdlow et al., 2016). Although memantine was found to impair learning as well as markers of perceptual and neural processing in some tasks (e.g., Rammsayer, 2001, 2006; Schugens et al., 1997), most studies found memantine to improve learning and to boost perceptual and neural processing (Golisch et al., 2017; Korostenskaja et al., 2007; Light et al., 2017; Swerdlow et al., 2016, 2020). As these previous studies typically used auditory stimuli, our results with visual stimuli add to a growing body of literature indicating that memantine can enhance sensory function, instead of decreasing it, and is the first study explicitly linking memantine to recurrent processing.

At this point we can only speculate why NMDA receptor blockade via memantine increased (rather than decreased) decoding accuracy in the condition measuring illusion-specific feedback. One speculative possibility is that NMDA receptor blockade may have reduced signal variability or entropy, resulting in more stereotypical neural processing, which may have improved classifier performance. The effect of memantine on the excitation/inhibition (E/I) balance in human participants with schizophrenia, as measured noninvasively by computing the aperiodic, 1/f profile of the EEG power spectrum (Molina et al., 2020), may provide one explanation. A flatter 1/f power spectrum is thought to reflect noisier signal processing. Changes in E/I balance have been linked to variations in states of consciousness (He et al., 2010; Nir et al., 2008), and may also affect decoding performance due to changes in neural noise. As mentioned in the Introduction, this may be related to memantine modulating processing at other pre- or post-synaptic receptors present at NMDA-rich synapses, specifically affecting extrasynaptic NMDA receptors in GABAergic cells (Milnerwood et al., 2010; Yao et al., 2022). Memantine's strong effect on extrasynaptic NMDA receptors in GABAergic cells leads to increases in ACh levels, which have been shown to increase firing rates and reduce firing rate variability in macaques (Herrero et al., 2013, 2008). This may represent a mechanism through which memantine (but not ketamine or the NMDA receptor antagonists used by Self and colleagues) could boost visually evoked activity. Clearly, further research is needed to elucidate the precise mechanisms by which memantine modulates feedback processing, how EEG decoding may relate to signal variability, and to explore its implications for perception.

However, our behavioral results show that improved decoding accuracy under memantine was not associated with boosted conscious perception. There are several possible reasons for this lack of behavioral correlate. For example, EEG decoding may be a more sensitive measure of the neural effects of memantine, in particular given that perceptual sensitivity may have been at floor

(masked condition, experiment 1) or ceiling (unmasked condition, experiment 1, and experiment 2). It is also possible that the present decoding results are merely epiphenomenal, not mapping onto functional improvements (e.g., Williams et al., 2007 [↗](#)). However, given that we found a tight link between these EEG decoding markers and behavioral performance in our previous work (Fahrenfort et al., 2017 [↗](#); Noorman et al., 2023 [↗](#)), it is possible that the effect of memantine was just too subtle to show up in changes in overt behavior. Similarly, although we cannot fully exclude that the effect of memantine reflected differential perceptual learning via memantine's impact on LTP, we did not find evidence for such an effect based on perceptual learning: dividing the experiments in four time bins did not reveal differences or changes over time in illusion detection between placebo and memantine (see [supplementary figure S6](#) [↗](#)).

Interestingly, we found an effect of memantine on illusion decoding only for temporally attended (experiment 1, AB task) and task-relevant Kanizsa stimuli (both experiments) that were presented without a backward mask. Although these results have to be interpreted with caution because the key interaction effects were not statistically significant, this suggests that object-based attention is crucial for memantine to exert its effects. Previous work has revealed that object-based attention primarily enhances neuronal firing rates compared to unattended objects, especially during the recurrent processing phase (Roelfsema, 2023 [↗](#); Thiele & Bellgrove, 2018 [↗](#)). Attention-related neural enhancement can be observed in many regions throughout cortex, e.g., in early and intermediate visual regions, as well as inferotemporal, parietal, and even frontal cortices (reviewed in Roelfsema, 2023 [↗](#)). It is hypothesized that when an object is attended, this facilitates the integration of its local elements and that it enables figure-ground segmentation to occur, crucial for the perception of coherent shapes and for the phenomenological properties of the Kanizsa illusion (Narayanan et al., 2013 [↗](#)). The observation that the effect of inhibiting NMDA receptors by memantine was only found for attended objects may suggest that memantine is most effective in conditions where recurrent processing is strongest. Previous work has shown that memantine also affects cholinergic receptors (Glasgow et al., 2017 [↗](#); Johnson et al., 2015 [↗](#)), known to be involved in attentional processes (Herrero et al., 2013 [↗](#), 2008 [↗](#); Thiele & Bellgrove, 2018 [↗](#)), which may partly explain why the effect of memantine might be dependent on attentional allocation. This preliminary finding deserves future investigation to further understand its mechanistic origin.

In conclusion, our study demonstrates that memantine exerts largely selective effects on feedback processing during human perceptual inference, and these effects are contingent upon the allocation of attention. These findings contribute to our understanding of the neural mechanisms underpinning consciousness and highlight the importance of considering attentional factors in the study of recurrent processing. Further research is needed to elucidate the precise mechanisms by which memantine modulates feedback processing and to explore its implications for our understanding of perceptual inference and consciousness.

Methods

Experiment 1

Participants

Thirty-three right-handed men with normal, or corrected-to-normal, vision were recruited after they had passed extensive physical and mental screening (for exclusion criteria see [Supplementary Material](#)). Three of them failed at the screening session's training task and did not proceed to the two experimental sessions (see [Procedure](#)). One dropped out during his first experimental session because of a headache, which was unrelated to memantine as he had taken the placebo. The remaining 29 participants (22±3 years old) were analyzed. The study was

approved by the Medical Ethical Committee of the Amsterdam Medical Centre (AMC) (NL64341.018.18) and the local ethics committee of the University of Amsterdam (2022-BC-14355). Participants gave informed consent and received monetary compensation.

Stimuli

The target stimulus set had a 2 Kanizsa illusion levels (absent/present) $\times$ 2 collinearity levels (absent/present) $\times$ 2 rotation levels (absent/present) design, resulting in eight stimuli (**Fig. 1A**). Three aligned Pac-Men induced the Kanizsa illusion. Collinearity was present when the stimuli's three other elements (the “two-legged white circles”) were aligned. The controls for both the illusion and collinearity were made by rotating their elements 90 degrees. Differences in local contrast were created by rotating the entire stimulus 180 degrees. The targets spanned 7.5 degrees by 8.3 degrees of visual angle. The distance between the three Pac-Man stimuli as well as between the three aligned two-legged white circles was 2.8 degrees of visual angle. Although neuronal responses to collinearity in primary visual cortex are most robust when this distance is smaller (Kapadia et al., 1995, 2000), longer-range horizontal connections between neurons with similar orientation selectivity can span distances corresponding to visual angles considerably greater than 2.8 degrees (Bosking et al., 1997; Stettler et al., 2002).

The distractor stimulus set was the same as the target stimulus set, with two exceptions. First, the distractors were red instead of black. Second, the distractors' six elements were rotated 180 degrees relative to the targets', so neither the illusion nor collinearity were present in the distractors. Masks consisted of six differently shaped elements, all capable of covering the targets' elements. Six masks were created by rotating the original mask five times by 60 degrees. They spanned 8.5 degrees by 9.1 degrees of visual angle. Finally, the design of the fixation cross was adapted from Thaler et al. (2013).

Task designs

The training dataset for our decoding analyses was obtained from a separate “template matching” task performed in the intake session. No drugs were administered during this “localizer” task. Here, every 900-1100 ms, a target stimulus was presented for 50 ms. Participants had to indicate whether the target matched the “template”, one of the eight target stimuli that was presented before the start of a block of trials. Each stimulus served as the template once. In short, the task design was 8 templates $\times$ (7 mismatches $\times$ 16 presentations + 1 match $\times$ 48 presentations), totaling 1280 trials, 3/10 of which were matches.

The main task was a rapid serial visual presentation (RSVP) task. Here, on every trial, participants were shown two targets (T1 and T2) within a RSVP of distractors (**Fig. 1B**). The targets and distractors had a stimulus onset asynchrony of 100 ms. T2 and each distractor were presented for 17 ms. With the aim of inducing an attentional blink, attending to T1 was made easier by presenting it for 67 ms. T1 was preceded by five distractors and T2 was followed by six distractors. For the first six participants, T2 and each distractor were presented for 10 ms and T1 for 40 ms (using the monitor's 100 Hz refresh rate). These durations were changed to make the task easier, so fewer volunteers would need to be excluded (see Procedure).

T2 visibility was manipulated in two ways. First, half the T2s were unmasked. The other half was immediately followed by three masks, each presented for 17 ms with an interstimulus interval of 0 ms. The three masks were selected randomly, but all differed from each other. Masking drastically reduced T2 visibility. Second, half the lags between the two targets were long: eight distractors (i.e., 900 ms). The other half was short: one or two distractors (i.e., 200 or 300 ms). The participant's short lag duration was determined beforehand, during the screening session (see Procedure). Short lags often induced an attentional blink (i.e., missing T2s). Following the RSVP sequence, participants indicated for each target whether it contained the Kanizsa illusion or not. Response screens lasted until the response. Response button mapping was counterbalanced within the task.

In short, each experimental session had an 8 T1 stimuli $\times$ 8 T2 stimuli $\times$ 2 masking levels (absent/present) $\times$ 2 lags (long/short) task design, resulting in 256 conditions. Each condition was presented four times, totaling 1024 trials.

For both the RSVP and the template matching task, participants were instructed to fixate on the fixation cross.

Procedure

The experiment consisted of three separate sessions conducted on different days: a four-hour screening session and two seven-hour experimental sessions. At the beginning of the screening session, the volunteers were screened for contraindications to using memantine. If there were no contraindications, participants proceeded to performing the template matching task with EEG and training for the RSVP task. To proceed to the experimental sessions, participants had to score well enough on this training task. Three participants were excluded for failing to achieve this. The training task was also used to discover the duration of the short lag that led to the lowest score to reliably induce an attentional blink. For all participants T2 performance at short lags was more than 5% lower than at long lags.

Memantine (20 mg) and the placebo were administered in different experimental sessions, separated by at least a week. These sessions were conducted using a randomized, double-blind crossover design. The order of drug administration (placebo or memantine first) was counterbalanced across participants. Participants were required to refrain from recreational drugs four weeks before the experiment and to refrain from alcohol two days before each experimental session. Experimental sessions started at 9:00 with the pill intake and briefly practicing the RSVP task. The actual RSVP task was performed when memantine reached peak plasma level, four hours after the pharmaceutical was administered. After this task, five minutes of resting state data were collected and another task, the local-global task (Bekinschtein et al., 2009 [link](#)), was performed. These latter two paradigms are beyond the scope of this paper.

Before pill intake (~9:00) and before (~13:00) and after (~15:30) the tasks, participants filled in the visual analogue scale (VAS) to measure the effects of memantine on subjective state (Bond & Lader, 1974 [link](#); Danion et al., 1989 [link](#)). Tasks were programmed in Presentation software (Neurobehavioral Systems) and displayed on a 23 inch, 60 Hz, 1920 $\times$ 1080 pixels monitor, viewed at an 80 cm distance with the use of a chin rest.

Behavioral analysis

To investigate behavioral performance on the RSVP task, we used perceptual sensitivity (d') and bias (criterion, see [supplementary figure S1 link](#)) derived from signal detection theory. The attentional blink requires T1 to be attended, so trials with incorrect T1s were excluded when T2 performance was investigated. To investigate the VAS measurements, the mean scores of subsets of the scales were calculated and taken as measures of alertness, contentedness, and calmness following Bond and Lader (1974 [link](#)) and of sedation following Danion et al. (1989 [link](#)). VAS scores of the two measurements after pill intake were averaged and corrected by calculating percentage change from the baseline measurement. For all behavioral analyses, repeated measures ANOVAs and double-sided (Bayesian) t-tests were used to test the differences between our conditions.

EEG recording and preprocessing

EEG was recorded at 1024 Hz using a 64 channel ActiveTwo system (BioSemi). Four electrooculographic (EOG) electrodes measured horizontal and vertical eye movements. The data were analyzed with MATLAB (MathWorks). For most of the preprocessing steps, EEGLAB was used (Delorme & Makeig, 2004 [link](#)). The data were re-referenced to the earlobes. Poor channels were interpolated. High-pass filtering can cause artifacts in decoding analyses, so for slow-drift removal

we used trial-masked robust detrending (van Driel et al., 2021 [DOI](#)). Each trial was epoched from -250 to 1000 ms relative to target onset. To improve our independent component analysis (ICA), baseline correction was applied using the whole epoch as baseline (Groppe et al., 2009 [DOI](#)). ICA was used to remove blinks. Blink components were removed manually. Baseline correction was applied, now using a -250 to 0 ms window relative to target onset. Trials with values outside of a -300 to 300 microvolts range were removed. We used an adapted version of FieldTrip's `ft_artifact_zvalue` function to detect and remove trials with muscle artifacts (Oostenveld et al., 2011 [DOI](#)). As in the behavioral analyses, T2s preceded by incorrect T1s were excluded. Finally, the data were downsampled to 128 Hz.

Multivariate pattern analysis

We used the Amsterdam Decoding and Modeling (ADAM) toolbox to perform multivariate pattern analyses (MVPAs) on the EEG data (Fahrenfort et al., 2018 [DOI](#)). Using a tenfold cross-validation scheme, we first analyzed the template matching task's data. We decoded each visual feature (e.g., the Kanizsa illusion) by training linear discriminant classifiers to discriminate between the feature's absence and presence based on the raw EEG activity [across all electrodes]. Individual participants' data were split into ten equal-sized folds after randomizing the task's trial order. A classifier was then trained on nine folds and tested on the tenth one, ensuring independence of the training and testing sets. This procedure was repeated until each fold served as the test set once. Classifier performance, here area under the receiver operating characteristic (AUC) (Hand & Till, 2001 [DOI](#)), was averaged across all ten iterations. We applied "diagonal decoding": classifier performance was calculated for every time sample and the classifiers were tested on the same time sample they had been trained on.

Each decoded visual feature showed a clear peak in classifier performance over time (**Fig. 1E** [DOI](#)). Many of our analyses focus on the time windows encompassing these peaks: from 78 to 94 ms for local contrast, 192-292 ms for collinearity, and 223-323 ms for the illusion (following the same procedure to determine the different time windows as in our previous work, see Noorman et al., 2023 [DOI](#)). To obtain topographic maps showing the neural sources of the classifier performance, we multiplied the classifier weights with the data covariance matrix, yielding covariance/class separability maps (Haufe et al., 2014 [DOI](#)). The covariance/class separability maps of the three time windows showed that the classifiers depended on occipitoparietal electrodes during peak decoding accuracy (**Fig. 2A** [DOI](#)). We restricted the decoding analyses to using only these electrodes to ensure that any effects we observed were not due to poor signal-to-noise ratio.

Next, classifiers were trained on the template matchings task's data and tested on each condition of the RSVP task's data. We averaged each visual feature's time window in the training set's raw EEG data to increase power. Classifiers were trained on these averages and tested on all time samples of the test set, resulting in "off-diagonal decoding" (King & Dehaene, 2014 [DOI](#)). To establish the putative markers for the different forms of neural processing, each visual feature was decoded in this manner. When decoding a visual feature, the average of the same feature's time window was used as the training set. Off-diagonal decoding allowed us to focus on the neural processes of interest. For completeness, we also performed a diagonal decoding version of this train-test scheme. Here, classifiers trained on a time sample of the template matching task's data were tested on the same time sample of the RSVP task's data. The diagonal decoding results were qualitatively similar to the off-diagonal ones (supplementary figures S2-4 [DOI](#)).

For the time samples from -100 to 700 ms relative to target onset, we used a double-sided *t*-test to evaluate whether classifier performance differed from chance and whether the memantine conditions' classifier performance differed from the placebo conditions'. We used cluster-based permutation testing (10,000 iterations at a threshold of 0.05) to correct for multiple comparisons

(Maris & Oostenveld, 2007 [↗](#)). As in the behavioral analyses, repeated measures ANOVAs and double-sided (Bayesian) *t*-tests—here applied to the time windows’ average classifier performance—were used to test the differences between our conditions.

Experiment 2

Participants

Thirty right-handed men with normal, or corrected-to-normal, vision were recruited after they had passed extensive physical and mental screening. Two of them dropped out after the first experimental session because of personal reasons unrelated to memantine. EEG recording failed during an experimental session of one participant. The remaining 27 participants (22±2 years old) were analyzed. The study was approved by the Medical Ethical Committee of the Amsterdam Medical Centre (AMC) and the local ethics committee of the University of Amsterdam. Participants gave informed consent and received monetary compensation.

Stimuli

Stimuli were the same as in experiment 1, except for a small change to the visual feature collinearity, for which we removed the “feet” from the two-legged white circles ([Fig. S5](#) [↗](#)).

Task designs

The training dataset for our decoding analyses was obtained from a separate “localizer” task, performed in all three sessions. In the intake, no drugs were administered. In the other sessions the localizer was performed well before memantine reached peak plasma levels. Here, in separate blocks of trials each visual feature was task-relevant. On every trial, a target stimulus was presented for 83 ms. Participants had to fixate on the fixation cross and indicate whether the current task-relevant feature (e.g., the Kanizsa illusion) was present or absent. The (first) response window started at target onset and ended after 900 ms. If no response was made, “Respond faster!” was presented until response. Trials ended with 0-200 ms jitter and at least 300 ms between response and jitter. The order of the task-relevant features was counterbalanced over participants and, because each participant performed the task multiple times, kept the same within participants. Response button mapping was counterbalanced within the task. In short, the task design was 3 task-relevant features × 8 target stimuli × 72 presentations, totaling 1728 trials. Participants completed the task three times in separate sessions, for a total of 5184 trials (1728 trials per session). All trials were used to build the classifiers.

The main task the participants performed was the “task-relevance” task ([Fig. 4A](#) [↗](#)), which was the same as the localizer task, except for the following changes. Half the trials were masked and on these trials, the target stimulus was presented for 33 ms and immediately followed by three masks, the same way as in the RSVP task. As in the localizer task each visual feature was at one point task-relevant and thus attended, while at another point task-irrelevant and thus less attended. The task also contained a base-rate manipulation, whose effects are beyond the scope of this paper. In one block of trials, the task-relevant feature was present on 75% of the trials, while on the next block, it was present on 25% of the trials, or vice versa. Overall, the task-relevant feature’s was equally often present and absent, which made it possible to ignore the base-rate manipulation for the purpose of the present analyses. In short, the task design was 3 task-relevant features × 8 target stimuli × 2 masking levels (absent/present) × 80 presentations, totaling 3840 trials.

To achieve masked performance of ~75% correct (making it more comparable to a typical attentional blink performance), we staircased mask contrast using the weighted up-down method (Kaernbach, 1991 [↗](#)). Staircasing was done when the Kanizsa illusion was task-relevant, because masking affected this feature more than the other two. Contrast levels ranged from 0 (black) to 255

(white). Mask contrast started at level 230. Each correct response made the task more difficult: masks got darker by downward step size S_{down} . Each incorrect response made the task easier: masks got lighter by upward step size S_{up} . Step sizes were determined by $S_{\text{up}} \times p = S_{\text{down}} \times (1 - p)$, where p was the desired accuracy of 0.75. The smallest step size was always 13 contrast levels. The staircase ended after 25 reversals (a mistake after a correct response, or a correct response after a mistake). The mask contrast that was used during the experimental sessions was the average contrast level of the last 20 reversals.

Procedure

The procedure was the same as for experiment 1, except for the following changes. During the screening session, if the volunteers had no contraindications to using memantine, they now proceeded to performing the localizer task and practicing the task-relevance task. This practice included our staircase procedure. The mask strength obtained here was used during the experimental sessions. To increase the power of the decoding analyses, the pill intake at the start of the experimental sessions was now immediately followed by another session of the localizer task. The localizer task was followed by briefly practicing the task-relevance task. The actual task-relevance task was performed four hours after the pharmaceutical was administered. No additional tasks were performed. The experiment was conducted in two different types of labs. For 18 participants, tasks were displayed on a 23 inch, 1920×1080 pixels monitor and for the other participants, on a 27 inch, 2560×1440 pixels monitor.

Behavioral analysis

Behavioral analysis was the same as for experiment 1. One of the baseline VAS forms of one participant was not filled in, so he was excluded from the VAS analysis.

EEG recording and preprocessing

EEG recording and preprocessing were the same as for experiment 1.

Multivariate pattern analysis

Multivariate pattern analysis was the same as for experiment 1, except for the three instances of the localizer task now together serving the role of the template matching task and the task-relevance task serving the role of the RSVP task. When decoding a visual feature (e.g., the Kanizsa illusion), not all of the localizer task's trials were used, only those during which the same visual feature was task-relevant. The tenfold cross-validation scheme was applied to the localizer task and each decoded visual feature again showed a clear peak in classifier performance over time (**Fig. 4D**). Here, the time windows encompassing these peaks were from 78 to 94 ms for local contrast, 247-347 ms for collinearity, and 217-317 ms for the illusion. Covariance/class separability maps again showed that the classifiers depended on occipitoparietal electrodes during peak decoding accuracy, so our decoding analyses were restricted to using only these electrodes.

Additional files

Supplementary information [↗](#)

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Reviewer #1 (Public review):

The authors investigate the function and neural circuitry of reentrant signals in visual cortex. Recurrent signaling is thought to be necessary to common types of perceptual experience that are defined by long-range relationships or prior expectation. Contour illusions - where perceptual objects are implied by stimuli characteristics - are a good example of this. The perception of these illusions is thought to emerge as recurrent signals from higher cortical areas feedback onto early visual cortex, to tell early visual cortex that it should be seeing object contours where none are actually present.

The authors test the involvement of reentrant cortical activity in this kind of perception using a drug challenge. Reentrance in visual cortex is thought to rely on NMDAR-mediated glutamate signalling. The authors accordingly employ an NMDA antagonist to stop this mechanism, looking for the effect of this manipulation on visually evoked activity recorded in EEG.

The motivating hypothesis for the paper is that NMDA antagonism should stop recurrent activity, and that this should degrade perceptual activity supporting perception of a contour illusion, but not other types of visual experience. Results in fact show the opposite. Rather than degrading cortical activity evoked by the illusion, memantine makes it more likely that machine learning classification of EEG will correctly infer the presence of the illusion.

On the face of it, this is confusing. But the paper does a good job of providing possible accounts based on specific details of neurochemical signalling and receptor populations.

I broadly find the paper interesting, graceful, and creative. The hypotheses are clear and compelling, the techniques for both manipulation of brain state and observation of that impact are cutting edge and well suited, and the paper draws clear and convincing conclusions that are made necessary by the results. The work sits at the very interesting crux of systems neuroscience, neuroimaging, and pharmacology.

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Reviewer #2 (Public review):

This study presents an important finding to the field interested in recurrent processing and the role of NMDA-receptors herein. The evidence for improved decoding under memantine is convincing, while some open questions remain to be followed up in future studies (the lack of a behavioural effect, why is decoding improved rather than decreased?). It is an excellent example of how an unexpected finding can generate novel research ideas to the mechanisms underlying recurrent processing, suggesting that the answer lies in the differences in the effects of ketamine and memantine, rather than their commonalities.

I would like to thank the authors for the great care they have taken in addressing my concerns. I think the revised manuscript is significantly easier to follow now that specific hypothesis have been formulated in the introduction, and the direction of the results is explicitly stated throughout the manuscript. I further appreciate the dampening of some of the claims that are not completely supported by the appropriate interactions.

I think the resulting manuscript is an incredibly exciting contribution to our understanding of NMDA-receptor function, and a great example of how an unexpected finding can raise questions that could potentially drive the field forward. It shows how NMDA's role in recurrent processing is much more complicate than previously assumed, and reveals that it is not the commonalities between memantine and ketamine that are important in

understanding recurrent processing, but rather the differences. I look forward to future studies that will target these differences.

Overall great job.

<https://doi.org/10.7554/eLife.100530.2.sa2>

Reviewer #3 (Public review):

Summary:

In this study, Stein and colleagues use a clever masking/attentional blink paradigm using Kanisza stimuli, coupled with EEG decoding and the NMDA antagonist memantine, to isolate putative neural markers of feedforward, lateral, and feedback processing.

In two elegant experiments, they show that memantine selectively influences EEG decoding of only illusory Kanisza surfaces (but not contour continuation or raw contrast), only when unmasked, only when attention is available (not when "blinked"), and only when task-relevant.

This neatly implicates NMDA receptors in the feedback mechanisms that are believed to be involved in inferring illusory Kanisza surfaces, and builds a difficult bridge between the large body of human perceptual experiments and pharmacological and neurophysiological work in animals.

Strengths:

Three key strengths of the paper are 1) its elegant and thorough experimental design, which includes internal replication of some key findings, and 2) the clear pattern of results across the full set of experiments, and 3) its clear writing and presentation of results.

The paper effectively reports a 4-way interaction, with memantine only influencing decoding of surfaces (1) that are unmasked (2), with attention available (3) and task-relevant (4). Nevertheless, the results are very clear, with a clear separation between null effects on other conditions and quite a strong (and thus highly selective) effect on this one intersection of conditions. This makes the pattern of findings very convincing.

Weaknesses:

Overall this is an impressive and important paper. However, to my mind there are two minor weaknesses.

First, despite its clear pattern of neural effects, there is no corresponding perceptual effect. Although the manipulation fits neatly within the conceptual framework, and there are many reasons for not finding such an effect (floor and ceiling effects, narrow perceptual tasks etc), this does leave open the possibility that the observation is entirely epiphenomenal, and that the mechanisms being recorded here are not actually causally involved in perception per se.

Second, although it is clear that there is an effect on decoding in this particular condition, what that means is not entirely clear - particularly since performance improves, rather than decreases. It should be noted here that improvements in decoding performance do not necessarily need to map onto functional improvements, and we should all be careful to remain agnostic about what is driving classifier performance. Here too, the effect of memantine on decoding might be epiphenomenal - unrelated to the information carried in the neural population, but somehow changing the balance of how that is electrically aggregated on the surface of the skull. *Something* is changing, but that might be a

neurochemical or electrical side-effect unrelated to actual processing (particularly since no corresponding behavioural impact is observed.)

Comments on revisions:

I think the authors responded fairly to my comments. Even if they weren't really able to add new insight into why behaviour didn't show the same effects as decoding, they discuss this in the revised text.

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Author response:

The following is the authors' response to the original reviews

Reviewer #1:

(1) I miss some treatment of the lack of behavioural correlate. What does it mean that memantine benefits EEG classification accuracy without improving performance? One possibility here is that there is an improvement in response latency, rather than perceptual sensitivity. Is there any hint of that in the RT results? In some sort of combined measure of RT and accuracy?

First, we would like to thank the reviewer for their positive assessment of our work and for their extremely helpful and constructive comments that helped to significantly improve the quality of our manuscript.

The reviewer rightly points out that, to our surprise, we did not obtain a correlate of the effect of memantine in our behavioral data, neither in the reported accuracy data nor in the RT data. We do not report RT results as participants were instructed to respond as accurately as possible, without speed pressure. We added a paragraph in the discussion section to point to possible reasons for this surprising finding:

“There are several possible reasons for this lack of behavioral correlate. For example, EEG decoding may be a more sensitive measure of the neural effects of memantine, in particular given that perceptual sensitivity may have been at floor (masked condition, experiment 1) or ceiling (unmasked condition, experiment 1, and experiment 2). It is also possible that the present decoding results are merely epiphenomenal, not mapping onto functional improvements (e.g., Williams et al., 2007). However, given that we found a tight link between these EEG decoding markers and behavioral performance in our previous work (Fahrenfort et al., 2017; Noorman et al., 2023), it is possible that the effect of memantine was just too subtle to show up in changes in overt behavior.”

(2) An explanation is missing, about why memantine impacts the decoding of illusion but not collinearity. At a systems level, how would this work? How would NMDAR antagonism selectively impact long-range connectivity, but not lateral connectivity? Is this supported by our understanding of laminar connectivity and neurochemistry in the visual cortex?

We have no straightforward or mechanistic explanation for this finding. In the revised discussion, we are highlighting this finding more clearly, and included some speculative explanations:

“The present effect of memantine was largely specific to illusion decoding, our marker of feedback processing, while collinearity decoding, our marker of lateral processing, was not (experiment 1) or only weakly (experiment 2) affected by memantine. We have no

straightforward explanation for why NMDA receptor blockade would impact inter-areal feedback connections more strongly than intra-areal lateral connections, considering their strong functional interdependency and interaction in grouping and segmentation processes (Liang et al., 2017). One possibility is that this finding reflects properties of our EEG decoding markers for feedback vs. lateral processing: for example, decoding of the Kanizsa illusion may have been more sensitive to the relatively subtle effect of our pharmacological manipulation, either because overall decoding was better than for collinearity or because NMDA receptor dependent recurrent processes more strongly contribute to illusion decoding than to collinearity decoding.”

(3) The motivating idea for the paper is that the NMDAR antagonist might disrupt the modulation of the AMPA-mediated glu signal. This is in line with the motivating logic for Self et al., 2012, where NMDAR and AMPAR efficacy in macaque V1 was manipulated via microinfusion. But this logic seems to conflict with a broader understanding of NMDA antagonism. NMDA antagonism appears to generally have the net effect of increasing glu (and ACh) in the cortex through a selective effect on inhibitory GABAergic cells (eg. Olney, Newcomer, & Farber, 1999). Memantine, in particular, has a specific impact on extrasynaptic NMDARs (that is in contrast to ketamine; Milnerwood et al, 2010, Neuron), and this type of receptor is prominent in GABA cells (eg. Yao et al., 2022, JoN). The effect of NMDA antagonists on GABAergic cells generally appears to be much stronger than the effect on glutamergic cells (at least in the hippocampus; eg. Grunze et al., 1996).

This all means that it's reasonable to expect that memantine might have a benefit to visually evoked activity. This idea is raised in the GD of the paper, based on a separate literature from that I mentioned above. But all of this could be better spelled out earlier in the paper, so that the result observed in the paper can be interpreted by the reader in this broader context.

To my mind, the challenging task is for the authors to explain why memantine causes an increase in EEG decoding, where microinfusion of an NMDA antagonist into V1 reduced the neural signal Self et al., 2012. This might be as simple as the change in drug... memantine's specific efficacy on extrasynaptic NMDA receptors might not be shared with whatever NMDA antagonist was used in Self et al. 2012. Ketamine and memantine are already known to differ in this way.

We addressed the reviewer’s comments in the following way. First, we bring up our (to us, surprising) result already at the end of the Introduction, pointing the reader to the explanation mentioned by the reviewer:

“We hypothesized that disrupting the reentrant glutamate signal via blocking NMDA receptors by memantine would impair illusion and possibly collinearity decoding, as putative markers of feedback and lateral processing, but would spare the decoding of local contrast differences, our marker of feedforward processing. To foreshadow our results, memantine indeed specifically affected illusion decoding, but enhancing rather than impairing it. In the Discussion, we offer explanations for this surprising finding, including the effect of memantine on extrasynaptic NMDA receptors in GABAergic cells, which may have resulted in boosted visual activity.”

Second, as outlined in the response to the first point by Reviewer #2, we are now clear throughout the title, abstract, and paper that memantine “improved” rather than “modulated” illusion decoding.

Third, and most importantly, we restructured and expanded the Discussion section to include the reviewer’s proposed mechanisms and explanations for the effect. We would like to thank the reviewer for pointing us to this literature. We also discuss the results of Self et al. (2012), specifically the distinct effects of the two NMDAR antagonists used in this study, more

extensively, and speculate that their effects may have been similar to ketamine and thus possibly opposite of memantine (for the feedback signal):

“Although both drugs are known to inhibit NMDA receptors by occupying the receptor’s ion channel and are thereby blocking current flow (Glasgow et al., 2017; Molina et al., 2020), the drugs have different actions at receptors other than NMDA, with ketamine acting on dopamine D2 and serotonin 5-HT2 receptors, and memantine inhibiting several subtypes of the acetylcholine (ACh) receptor as well as serotonin 5HT3 receptors. Memantine and ketamine are also known to target different NMDA receptor subpopulations, with their inhibitory action displaying different time courses and intensity (Glasgow et al., 2017; Johnson et al., 2015). Blockade of different NMDA receptor subpopulations can result in markedly different and even opposite results. For example, Self and colleagues (2012) found overall reduced or elevated visual activity after microinfusion of two different selective NMDA receptor antagonists (2-amino-5phosphonovalerate and ifendprodil) in macaque primary visual cortex. Although both drugs impaired the feedback-related response to figure vs. ground, similar to the effects of ketamine (Meuwese et al., 2013; van Loon et al., 2016) such opposite effects on overall activity demonstrate that the effects of NMDA antagonism strongly depend on the targeted receptor subpopulation, each with distinct functional properties.”

Finally, we link these differences to the potential mechanism via GABAergic neurons:

“As mentioned in the Introduction, this may be related to memantine modulating processing at other pre- or post-synaptic receptors present at NMDA-rich synapses, specifically affecting extrasynaptic NMDA receptors in GABAergic cells (Milnerwood et al, 2010; Yao et al., 2022). Memantine’s strong effect on extrasynaptic NMDA receptors in GABAergic cells leads to increases in ACh levels, which have been shown to increase firing rates and reduce firing rate variability in macaques (Herrero et al., 2013, 2008). This may represent a mechanism through which memantine (but not ketamine or the NMDA receptor antagonists used by Self and colleagues) could boost visually evoked activity.”

(4) The paper's proposal is that the effect of memantine is mediated by an impact on the efficacy of reentrant signaling in visual cortex. But perhaps the best-known impact of NMDAR manipulation is on LTP, in the hippocampus particularly but also broadly.

Perception and identification of the kanisza illusion may be sensitive to learning (eg. Maertens & Pollmann, 2005; Gellatly, 1982; Rubin, Nakayama, Shapley, 1997); what argues against an account of the results from an effect on perceptual learning? Generally, the paper proposes a very specific mechanism through which the drug influences perception. This is motivated by results from Self et al 2012 where an NMDA antagonist was infused into V1. But oral memantine will, of course, have a whole-brain effect, and some of these effects are well characterized and - on the surface - appear as potential sources of change in illusion perception. The paper needs some treatment of the known ancillary effects of diffuse NMDAR antagonism to convince the reader that the account provided is better than the other possibilities.

We cannot fully exclude an effect based on perceptual learning but consider this possibility highly unlikely for several reasons. First, subjects have performed more than a thousand trials in a localizer session before starting the main task (in experiment 2 even more than two thousand) containing the drug manipulation. Therefore, a large part of putative perceptual learning would have already occurred before starting the main experiment. Second, the main experiment was counterbalanced across drug sessions, so half of the participants first performed the memantine session and then the placebo session, and the other half of the subjects the other way around. If memantine would have improved perceptual learning in our experiments, one may actually expect to observe improved decoding in the placebo session and not in the memantine session. If memantine would have facilitated perceptual

learning during the memantine session, the effect of that facilitated perceptual learning would have been most visible in the placebo session following the memantine session. Because we observed improved decoding in the memantine session itself, perceptual learning is likely not the main explanation for these findings. Third, perceptual learning is known to occur for several stimulus dimensions (e.g., orientation, spatial frequency or contrast). If these findings would have been driven by perceptual learning one would have expected to see perceptual learning for all three features, whereas the memantine effects were specific to illusion decoding. Especially in experiment 2, all features were equally often task relevant and in such a situation one would've expected to observe perceptual learning effects on those other features as well.

To further investigate any potential role of perceptual learning, we analyzed participants' performance in detecting the Kanizsa illusion over the course of the experiments. To investigate this, we divided the experiments' trials into four time bins, from the beginning until the end of the experiment. For the first experiment's first target (T1), there was no interaction between the factors bin and drug (memantine/placebo; $F_{3,84}=0.89$, $P=0.437$; Figure S6A). For the second target (T2), we performed a repeated-measures ANOVA with the factors bin, drug, T1-T2 lag (short/long), and masks (present/absent). There was only a trend towards a bin by drug interaction ($F_{3,84}=2.57$, $P=0.064$; Figure S6B), reflecting worse performance under memantine in the first three bins and slightly better performance in the fourth bin. The other interactions that include the factors bin and drug factors were not significant (all $P>0.117$). For the second experiment, we performed a repeated-measures ANOVA with the factors bin, drug, masks, and task-relevant feature (local contrast/collinearity/illusion). None of the interactions that included the bin and drug factors were significant (all $P>0.219$; Figure S6C). Taken together, memantine does not appear to affect Kanizsa illusion detection performance through perceptual learning. Finally, there was no interaction between the factors bin and task-relevant feature ($F_{6,150}=0.76$, $P=0.547$; Figure S6D), implying there is no perceptual learning effect specific to Kanizsa illusion detection. We included these analyses in our revised Supplement as Fig. S6.

(5) The cross-decoding approach to data analysis concerns me a little. The approach adopted here is to train models on a localizer task, in this case, a task where participants matched a kanisza figure to a target template (E1) or discriminated one of the three relevant stimuli features (E2). The resulting model was subsequently employed to classify the stimuli seen during separate tasks - an AB task in E1, and a feature discrimination task in E2. This scheme makes the localizer task very important. If models built from this task have any bias, this will taint classifier accuracy in the analysis of experimental data. My concern is that the emergence of the kanisza illusion in the localizer task was probably quite salient, respective to changes in stimuli rotation or collinearity. If the model was better at detecting the illusion to begin with, the data pattern - where drug manipulation impacts classification in this condition but not other conditions - may simply reflect model insensitivity to non-illusion features.

I am also vaguely worried by manipulations implemented in the main task that do not emerge in the localizer - the use of RSVP in E1 and manipulation of the base rate and staircasing in E2. This all starts to introduce the possibility that localizer and experimental data just don't correspond, that this generates low classification accuracy in the experimental results and ineffective classification in some conditions (ie. when stimuli are masked; would collinearity decoding in the unmasked condition potentially differ if classification accuracy were not at a floor? See Figure 3c upper, Figure 5c lower).

What is the motivation for the use of localizer validation at all? The same hypotheses can be tested using within-experiment cross-validation, rather than validation from a model built on localizer data. The argument may be that this kind of modelling will necessarily employ a smaller dataset, but, while true, this effect can be minimized at the expense of

computational cost - many-fold cross-validation will mean that the vast majority of data contributes to model building in each instance.

It would be compelling if results were to reproduce when classification was validated in this kind of way. This kind of analysis would fit very well into the supplementary material.

We thank the reviewer for this excellent question. We used separate localizers for several reasons, exactly to circumvent the kind of biases in decoding that the reviewer alludes to. Below we have detailed our rationale, first focusing on our general rationale and then focusing on the decisions we made in designing the specific experiments.

Using a localizer task in the design of decoding analysis offers several key advantages over relying solely on k-fold cross-validation within the main task:

(1) Feature selection independence and better generalization: A separate localizer task allows for independent feature selection, ensuring that the features used for decoding are chosen without bias from the main task data. Specifically, the use of a localizer task allows us to determine the time-windows of interest independently based on the peaks of the decoding in the localizer. This allows for a better direct comparison between the memantine and placebo conditions because we can isolate the relevant time windows outside a drug manipulation. Further, training a classifier on a localizer task and testing it on a separate experimental task assesses whether neural representations generalize across contexts, rather than simply distinguishing conditions within a single dataset. This supports claims about the robustness of the decoded information.

(2) Increased sensitivity and interpretability: The localizer task can be designed specifically to elicit strong, reliable responses in the relevant neural patterns. This can improve signal-to-noise ratio and make it easier to interpret the features being used for decoding in the test set. We facilitate this by having many more trials in the localizer tasks (1280 in E1 and 5184 in E2) than in the separate conditions of the main task, in which we would have to do k-folding (e.g., 2, mask, x 2 (lag) design in E1 leaves fewer than 256 trials, due to preprocessing, for specific comparisons) on very low trial numbers. The same holds for experiment 2 which has a 2x3 design, but also included the base-rate manipulation. Finally, we further facilitate sensitivity of the model by having the stimuli presented at full contrast without any manipulations of attention or masking during the localizer, which allows us to extract the feature specific EEG signals in the most optimal way.

(3) Decoupling task-specific confounds: If decoding is performed within the main task using k-folding, there is a risk that task-related confounds (e.g., motor responses, attention shifts, drug) influence decoding performance. A localizer task allows us to separate the neural representation of interest from these task-related confounds.

Experiment 1

In experiment 1, the Kanizsa was always task relevant in the main experiment in which we employed the pharmacological manipulation. To make sure that the classifiers were not biased towards Kanizsa figures from the start (which would be the case if we would have done k-folding in the main task), we used a training set in which all features were equally relevant for task performance. As can be seen in figure 1E, which plots the decoding accuracies of the localizer task, illusion decoding as well as rotation decoding were equally strong, whereas collinearity decoding was weaker. It may be that the Kanizsa illusion was quite salient in the localizer task, which we can't know at present, but it was at least less salient and relevant than in the main task (where it was the only task-relevant feature). Based on the localizer decoding results one could argue that the rotation dimension and illusion dimension were most salient, because the decoding was highest for these dimensions. Clearly the model was not insensitive to nonillusory features. The localizer task of experiment 2

reveals that collinearity decoding tends to be generally lower, even when that feature is task relevant.

Experiment 2

In experiment 2, the localizer task and main task were also similar, with three exceptions: during the localizer task no drug was active, and no masking and no base rate manipulation were employed. To make sure that the classifier was not biased towards a certain stimulus category (due to the bias manipulation), e.g. the stimulus that is presented most often, we used a localizer task without this manipulation. As can be seen in figure 4D decoding of all the features was highly robust, also for example for the collinearity condition. Therefore the low decoding that we observe in the main experiment cannot be due to poor classifier training or feature extraction in the localizer. We believe this is actually an advantage instead of a disadvantage of the current decoding protocol.

Based on the rationale presented above we are uncomfortable performing the suggested analyses using a k-folding approach in the main task, because according to our standards the trial numbers are too low and the risk that these results are somehow influenced by task specific confounds cannot be ruled out.

Line 301 - 'Interestingly, in both experiments the effect of memantine... was specific to... stimuli presented without a backward mask.' This rubs a bit, given that the mask broadly disrupted classification. The absence of memantine results in masked results may simply be a product of the floor ... some care is needed in the interpretation of this pattern.

In the results section of experiment 1, we added:

“While the interaction between masking and memantine only approached significance ($P=0.068$), the absence of an effect of memantine in the masked condition could reflect a floor effect, given that illusion decoding in the masked condition was not significantly better than chance.”

While floor is less likely to account for the absence of an effect in the masked condition in experiment 2, where illusion decoding in the masked condition was significantly above chance, it is still possible that to obtain an effect of memantine, decoding accuracy needed to be higher. We therefore also added here:

“For our time window-based analyses of illusion decoding, the specificity of the memantine effect to the unmasked condition was supported by a significant interaction between drug and masking (note, however, given overall much lower decoding accuracy in the masked condition, the lack of a memantine effect could reflect a floor effect).”

In the discussion, we changed the sentence to read “...the effect of memantine on illusion decoding *tended* to be specific to attended, task-relevant stimuli presented without a backward mask.”

Line 441 - What were the contraindications/exclusion parameters for the administration of memantine?

Thanks for spotting this. We have added the relevant exclusion criteria in the revised version of the supplement. See also below.

- Allergy for memantine or one of the inactive ingredients of these products;
- (History of) psychiatric treatment;
- First-degree relative with (history of) schizophrenia or major depression;

- (History of) clinically significant hepatic, cardiac, obstructive respiratory, renal, cerebrovascular, metabolic or pulmonary disease, including, but not limited to fibrotic disorders;
- Claustrophobia;
- Regular usage of medicines (antihistamines or occasional use of paracetamol);
- (History of) neurological disease;
- (History of) epilepsy;
- Abnormal hearing or (uncorrected) vision;
- Average use of more than 15 alcoholic beverages weekly;
- Smoking
- History of drug (opiate, LSD, (meth)amphetamine, cocaine, solvents, cannabis, or barbiturate) or alcohol dependence;
- Any known other serious health problem or mental/physical stress;
- Used psychotropic medication, or recreational drugs over a period of 72 hours prior to each test session,
- Used alcohol within the last 24 hours prior to each test session;
- (History of) pheochromocytoma.
- Narrow-angle glaucoma;
- (History of) ulcer disease;
- Galactose intolerance, Lapp lactase deficiency or glucosegalactose malabsorption.
- (History of) convulsion;

Line 587 - The localizer task used to train the classifier in E2 was collected in different sessions. Was the number of trials from separate sessions ultimately equal? The issue here is that the localizer might pick up on subtle differences in electrode placement. If the test session happens to have electrode placement that is similar to the electrode placement that existed for a majority of one condition of the localizer... this will create bias. This is likely to be minor, but machine classifiers really love this kind of minor confound.

Indeed, the trial counts in the separate sessions for the localizer in E2 were equal. We have added that information to the methods section.

Experiment 1: 1280 trials collected during the intake session.

In experiment 2: 1728 trials were collected per session (intake, and 2 drug sessions), so there were 5184 trials across three sessions.

Reviewer #2:

To start off, I think the reader is being a bit tricked when reading the paper. Perhaps my priors are too strong, but I assumed, just like the authors, that NMDA-receptors would disrupt recurrent processing, in line with previous work. However, due to the continuous

use of the ambiguous word 'affected' rather than the more clear increased or perturbed recurrent processing, the reader is left guessing what is actually found. That's until they read the results and discussion finding that decoding is actually improved. This seems like a really big deal, and I strongly urge the authors to reword their title, abstract, and introduction to make clear they hypothesized a disruption in decoding in the illusion condition, but found the opposite, namely an increase in decoding. I want to encourage the authors that this is still a fascinating finding.

We thank the reviewer for the positive assessment of our manuscript, and for many helpful comments and suggestions.

We changed the title, abstract, and introduction in accordance with the reviewer's comment, highlighting that “memantine [...] improves decoding” and “enhances recurrent processing” in all three sections. We also changed the heading of the corresponding results section to “Memantine selectively improves decoding of the Kanizsa illusion”.

Apologies if I have missed it, but it is not clear to me whether participants were given the drug or placebo during the localiser task. If they are given the drug this makes me question the logic of their analysis approach. How can one study the presence of a process, if their very means of detecting that process (the localiser) was disrupted in the first place? If participants were not given a drug during the localiser task, please make that clear. I'll proceed with the rest of my comments assuming the latter is the case. But if the former, please note that I am not sure how to interpret their findings in this paper.

Thanks for asking this, this was indeed unclear. In experiment 1 the localizer was performed in the intake session in which no drugs were administered. In the second experiment the localizer was performed in all three sessions with equal trial numbers. In the intake session no drugs were administered. In the other two sessions the localizer was performed directly after pill intake and therefore the memantine was not (or barely) active yet. We started the main task four hours after pill intake because that is the approximate peak time of memantine. Note that all three localizer tasks were averaged before using them as training set. We have clarified this in the revised manuscript.

The main purpose of the paper is to study recurrent processing. The extent to which this study achieves this aim is completely dependent to what extent we can interpret decoding of illusory contours as uniquely capturing recurrent processing. While I am sure illusory contours rely on recurrent processing, it does not follow that decoding of illusory contours capture recurrent processing alone. Indeed, if the drug selectively manipulates recurrent processing, it's not obvious to me why the authors find the interaction with masking in experiment 2. Recurrent processing seems to still be happening in the masked condition, but is not affected by the NMDA-receptor here, so where does that leave us in interpreting the role of NMDA-receptors in recurrent processing? If the authors can not strengthen the claim that the effects are completely driven by affecting recurrent processing, I suggest that the paper will shift its focus to making claims about the encoding of illusory contours, rather than making primary claims about recurrent processing.

We indeed used illusion decoding as a marker of recurrent processing. Clearly, such a marker based on a non-invasive and indirect method to record neural activity is not perfect. To directly and selectively manipulate recurrent processing, invasive methods and direct neural recordings would be required. However, as explained in the revised Introduction,

“In recent work we have validated that the decoding profiles of these features of different complexities at different points in time, in combination with the associated topography, can

indeed serve as EEG markers of feedforward, lateral and recurrent processes (Fahrenfort et al., 2017; Noorman et al., 2023)."

The timing and topography of the decoding results of the present study were consistent with our previous EEG decoding studies (Fahrenfort et al., 2017; Noorman et al., 2023). This validates the use of these EEG decoding signatures as (imperfect) markers of distinct neural processes, and we continue to use them as such. However, we expanded the discussion section to alert the reader to the indirect and imperfect nature of these EEG decoding signatures as markers of distinct neural processes: "Our approach relied on using EEG decoding of different stimulus features at different points in time, together with their topography, as markers of distinct neural processes. Although such non-invasive, indirect measures of neural activity cannot provide direct evidence for feedforward vs. recurrent processes, the timing, topography, and susceptibility to masking of the decoding signatures obtained in the present study are consistent with neurophysiology (e.g., Bosking et al., 1997; Kandel et al., 2000; Lamme & Roelfsema, 2000; Lee & Nguyen, 2001; Liang et al., 2017; Pak et al., 2020), as well as with our previous work (Fahrenfort et al., 2017; Noorman et al., 2023)."

The reviewer is also concerned about the lack of effect of memantine on illusion decoding in the masked condition in experiment 2. In our view, the strong effect of masking on illusion decoding (both in absolute terms, as well as when compared to its effect on local contrast decoding), provides strong support for our assumption that illusion decoding represents a marker of recurrent processing. Nevertheless, as the reviewer points out, weak but statistically significant illusion decoding was still possible in the masked condition, at least when the illusion was task-relevant. As the reviewer notes, this may reflect residual recurrent processing during masking, a conclusion consistent with the relatively high behavioral performance despite masking ($d' > 1$). However, rather than invalidating the use of our EEG markers or challenging the role of NMDA-receptors in recurrent processing, this may simply reflect a floor effect. As outlined in our response to reviewer #1 (who was concerned about floor effects), in the results section of experiment 1, we added:

"While the interaction between masking and memantine only approached significance ($P=0.068$), the absence of an effect of memantine in the masked condition could reflect a floor effect, given that illusion decoding in the masked condition was not significantly better than chance."

And for experiment 1:

"For our time window-based analyses of illusion decoding, the specificity of the memantine effect to the unmasked condition was supported by a significant interaction between drug and masking (note, however, given overall much lower decoding accuracy in the masked condition, the lack of a memantine effect could reflect a floor effect)."

An additional claim is being made with regards to the effects of the drug manipulation. The authors state that this effect is only present when the stimulus is 1) consciously accessed, and 2) attended. The evidence for claim 1 is not supported by experiment 1, as the masking manipulation did not interact in the cluster-analyses, and the analyses focussing on the peak of the timing window do not show a significant effect either. There is evidence for this claim coming from experiment 2 as masking interacts with the drug condition. Evidence for the second claim (about task relevance) is not presented, as there is no interaction with the task condition. A classical error seems to be made here, where interactions are not properly tested. Instead, the presence of a significant effect in one condition but not the other is taken as sufficient evidence for an interaction, which is not appropriate. I therefore urge the authors to dampen the claim about the importance of attending to the decoded features. Alternatively, I suggest the authors run their interactions of interest on the time-courses and conduct the appropriate clusterbased analyses.

We thank the reviewer for pointing out the importance of key interaction effects. Following the reviewer's suggestion, we dampened our claims about the role of attention. For experiment 1, we changed the heading of the relevant results section from "Memantine's effect on illusion decoding requires attention" to "The role of consciousness and attention in memantine's effect on illusion decoding", and we added the following in the results section:

"Also our time window-based analyses showed a significant effect of memantine only when the illusion was both unmasked and presented outside the AB ($t_{28} = -2.76$, $P = 0.010$, $BF_{10} = 4.53$; Fig. 3F). Note, however, that although these post-hoc tests of the effect of memantine on illusion decoding were significant, for our time window-based analyses we did not obtain a statistically significant interaction between the AB and memantine, and the interaction between masking and memantine only approached significance ($P = 0.068$). Thus, although these memantine effects were slightly less robust than for T1, probably due to reduced trial counts, these results point to (but do not conclusively demonstrate) a selective effect of memantine on illusion-related feedback processing that depends on the availability of attention. In addition to the lack of the interaction effect, another potential concern..."

For experiment 2, we added the following in the results section:

"Note that, for our time window-based analyses of illusion decoding, although the specificity of the memantine effect to the unmasked condition was supported by a significant interaction between drug and masking, we did not obtain a statistically significant interaction between memantine and task-relevance. Thus, although the memantine effect was significant only when the illusion was unmasked and task-relevant, just like for the effect of temporal attention in experiment 1, these results do not conclusively demonstrate a selective effect of memantine that depends attention (task-relevance)."

In the discussion, we toned down claims about memantine's effects being specific to attended conditions, we are highlighting the "preliminary" nature of these findings, and we are now alerting the reader explicitly to be careful with interpreting these effects, e.g.:

"Although these results have to be interpreted with caution because the key interaction effects were not statistically significant, ..."

How were the length of the peak-timing windows established in Figure 1E? My understanding is that this forms the training-time window for the further decoding analyses, so it is important to justify why they have different lengths, and how they are determined. The same goes for the peak AUC time windows for the interaction analyses. A number of claims in the paper rely on the interactions found in these posthoc analyses, so the 223- to 323 time window needs justification.

Thanks for this question. The length of these peak-timing windows is different because the decoding of rotation is temporarily very precise and short-lived, whereas the decoding of the other features last much longer and is more temporally variable. In fact, we have followed the same procedure as in a previously published study (Noorman et al., eLife 2025) for defining the peak-timing and length of the windows. We followed the same procedure for both experiments reported in this paper, replicating the crucial findings and therefore excluding the possibility that these findings are in any way dependent on the time windows that are selected. We have added that information to the revised version of the manuscript.

Reviewer #3:

First, despite its clear pattern of neural effects, there is no corresponding perceptual effect. Although the manipulation fits neatly within the conceptual framework, and there are many reasons for not finding such an effect (floor and ceiling effects, narrow

perceptual tasks, etc), this does leave open the possibility that the observation is entirely epiphenomenal, and that the mechanisms being recorded here are not actually causally involved in perception per se.

We thank the reviewer for the positive assessment of our work. The reviewer rightly points out that, to our surprise, we did not obtain a correlate of the effect of memantine in our behavioral data. We agree with the possible reasons for the absence of such an effect highlighted by the reviewer, and expanded our discussion section accordingly:

“There are several possible reasons for this lack of behavioral correlate. For example, EEG decoding may be a more sensitive measure of the neural effects of memantine, in particular given that perceptual sensitivity may have been at floor (masked condition, experiment 1) or ceiling (unmasked condition, experiment 1, and experiment 2). It is also possible that the present decoding results are merely epiphenomenal, not mapping onto functional improvements (e.g., Williams et al., 2007). However, given that in our previous work we found a tight link between these EEG decoding markers and behavioral performance (Fahrenfort et al., 2017; Noorman et al., 2023), it is possible that the effect of memantine in the present study was just too subtle to show up in changes in overt behavior.”

*Second, although it is clear that there is an effect on decoding in this particular condition, what that means is not entirely clear - particularly since performance improves, rather than decreases. It should be noted here that improvements in decoding performance do not necessarily need to map onto functional improvements, and we should all be careful to remain agnostic about what is driving classifier performance. Here too, the effect of memantine on decoding might be epiphenomenal - unrelated to the information carried in the neural population, but somehow changing the balance of how that is electrically aggregated on the surface of the skull. *Something* is changing, but that might be a neurochemical or electrical side-effect unrelated to actual processing (particularly since no corresponding behavioural impact is observed.)*

We would like to refer to our reply to the previous point, and we would like to add that in our previous work (Fahrenfort et al., 2017; Noorman et al., 2023) similar EEG decoding markers were often tightly linked to changes in behavioral performance. This indicates that these particular EEG decoding markers do not simply reflect some sideeffect not related to neural processing. However, as stated in the revised discussion section, “it is possible that the effect of memantine in the present study was just too subtle to show up in changes in overt behavior.”

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